Aman Katyal

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Education

Purdue University — West Lafayette, IN
Master of Science in Electrical and Computer Engineering — Expected May 2028

Purdue University — West Lafayette, IN
Bachelor of Science in Computer Engineering | GPA: 3.91, Dean's List and Semester Honors — Expected May 2027

Experience

ASIC and Firmware Design Intern — May 2026 – Aug 2026
Hewlett Packard Enterprise — Roseville, CA

- Rebuilt an internal microcode compilation pipeline in Python with a defined grammar, AST validator, and CLI toolchain targeting on-chip instructions
- Parsed 10 GB of pre-silicon simulation trace output in under 2 minutes into cycle-indexed DuckDB schemas, isolating pipeline stalls and buffer bottlenecks
- Developed a binary disassembler that converts compiled microcode into a readable assembly representation, cutting manual debug turnaround time for architecture and RTL teams

Firmware and Hardware Lead — May 2026 – Present
Purdue IEEE ROV Team — West Lafayette, IN

- Migrated vehicle communication from point-to-point SPI to a multi-node CAN FD bus across 3 STM32 microcontrollers and led firmware bring-up across a multi-board stack
- Architected a cross-platform `rov` CLI (Python + `rov.toml`) that abstracts CMake and Ninja setup, ST-Link flashing, and serial monitoring into one-word build, flash, and monitor commands
- Built a software-in-the-loop harness (mocked multi-node CAN FD bus with fault injection) spanning 25 CTest suites on a hardware abstraction layer (HAL) and board support package (BSP), enforced as a blocking GitHub Actions CI gate
- Designed a 4-layer KiCad PCB for hardware-in-the-loop validation and benchmarked an RP2350B against STM32

Embedded Software Engineer — Aug 2025 – Present
Purdue System on Chip Extension Technologies (SoCET) — West Lafayette, IN

- Cut dynamic stack memory traffic 87.9% (680 → 82 ops) in a RISC-V AI-accelerator compiler by fixing a frontend mem2reg bailout and expanding the scalar register pool from 19 to 26 registers (+37%), eliminating stack spills from all core kernels
- Reduced benchmark-suite execution 13.8% (3,656 → 3,153 cycles) and retired instructions 13.7% across 12 AI kernels via loop-depth-weighted register allocation, loop-invariant code motion, and redundant-load elimination; compiled kernels now beat hand-written assembly (92 vs 103 cycles)
- Built a dual-baseline validation framework for the AI-accelerator compiler (roofline and very long instruction word (VLIW) slot-density metrics) with 11/11 kernels matching NumPy goldens and 12/12 ISA tests passing
- Ported TinyUSB to a separate custom RISC-V SoC, implementing a bare-metal USB 1.1 host controller driver with a platform-level interrupt controller (PLIC)-driven interrupt service routine and DMA data paths

Projects

ROV Autonomous Profiling Module | RP2040, PID, LoRa, OTA — Jan 2026 – May 2026

- Wrote bare-metal C firmware on an RP2040 including custom I2C sensor drivers with 256x oversampling and bus recovery, for a buoyancy engine that holds depth within 5 cm
- Implemented a 10 Hz feedforward PID controller and tuned it against logged depth data, extending neutral-buoyancy hold stability during profiling
- Cut firmware refresh time from 8 minutes to under 20 seconds with a dual-slot over-the-air (OTA) bootloader using bsdiff differential patching over LoRa

Harry Potter Wand Duel | RP2350, TinyML, PIO, DMA, Custom PCB — Aug 2025 – Dec 2025

- Built a dual-core RP2350 system with a jitter-free 74 Hz IMU sampling loop on core 0, and TensorFlow Lite Micro gesture inference at 96% accuracy on core 1 linked via SIO FIFOs and mutexes
- Wrote PIO-driven 38 kHz NEC IR and 16x16 RGB LED matrix drivers with custom haptic feedback for a custom PCB

Skills

Languages: C (Bare-Metal), C++, Python, RISC-V Assembly, SystemVerilog
Embedded: ARM Cortex-M, STM32, RP2040, RP2350, Bootloaders, DMA, ISRs, PID Control, TinyML, PIO
Protocols and Tools: SPI, I2C, UART, CAN FD, USB 1.1 (Host), KiCad, Git, CMake, GitHub Actions